British Food Journal

The emerging research frontier of food analytics: artificial intelligence, consumer analytics, smart food systems and knowledge-driven intelligence

Received 18 July 2026
Revised 22 August 2026
12 September 2026
19 September 2026
Accepted 30 September 2026

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Abstract

Purpose – This study examines the emerging research frontier of Food Analytics between 2016 and mid-2026. It identifies descriptors that show sustained novelty and growth, analyses how they combine into emergent thematic clusters, and synthesizes these clusters into broader research dimensions that shape the field's current research frontier.

Design/methodology/approach – The study adopts a technological-intelligence design based on bibliometric tech mining and analyzes 1,967 Scopus-indexed publications using descriptor normalization, Porter's emergence indicators, document-level co-word analysis, and Louvain community detection. Emergent descriptors were identified using temporal growth and novelty criteria, clustered by co-occurrence, and interpreted comparatively to derive higher-order analytical dimensions.

Findings – The analysis identified 111 emergent descriptors organized into five Louvain communities after a sensitivity analysis of the emergence-ratio threshold and evaluation of cluster coverage and interpretability. The five-community solution was synthesized into four substantive dimensions: intelligent food quality and composition analytics; predictive and operational food-system analytics; personalized nutrition and consumer analytics; and semantic, generative, and knowledge-driven food analytics. The difference between the five communities and four dimensions reflects the integration of a predominantly methodological community across the substantive dimensions in which its methods are applied.

Originality/value – The study moves beyond technology-specific reviews by identifying emergence at the descriptor level and integrating overlapping clusters into a coherent framework of research dimensions. It provides a forward-looking account of how Food Analytics is expanding and clarifies the technological, methodological, and application areas most likely to shape its next stage of development.

Keywords Emerging research trends, Technology intelligence, Research frontiers, Smart food systems, AI in food systems

Paper type Research article

1. Introduction

Food Analytics is a developing research field that organizes and interprets food-related evidence using computational, instrumental, and data-intensive approaches. Its distinctive feature is not any single technology, but the integration of diverse data sources—such as molecular, spectral, sensory, operational, nutritional, and consumer data—into analytical systems that reveal patterns, relationships, and emerging applications within food contexts (Bordonali *et al.*, 2019). In this sense, Food Analytics provides a common analytical space in which methods from Artificial Intelligence (AI), spectroscopy, chemistry, bioinformatics, and data science are combined to generate food-specific insight across scientific and applied settings. Conceptually, the domain is therefore delimited not by the presence of a particular technology, but by the substantive integration of food-related data or evidence with analytical or computational methods to generate knowledge or decision support specific to food-system problems. Studies in which food is merely contextual or analytical techniques are used without a substantive contribution to understanding or decision-making within food systems fall outside this boundary.

British Food Journal
© Emerald Publishing Limited
e-ISSN: 1758-4108
p-ISSN: 0007-070X
DOI 10.1108/BFJ-07-2026-1291

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This study examines the emerging research frontier of Food Analytics by identifying descriptors exhibiting sustained novelty and growth between 2016 and mid-2026 and analyzing how they combine into broader thematic dimensions. The study takes a technological-intelligence perspective, treating bibliographic records as structured science-and-technology information from which emerging R&D signals and trajectories can be computationally identified. By integrating Porter's emergence indicators, co-word analysis, and Louvain community detection, the study reveals the principal dimensions currently shaping Food Analytics and provides a structured synthesis of the field's emerging technological and research directions.

The remainder of the article is structured as follows. Section 2 reviews the literature and positions the study. Section 3 presents the research design, data preparation, and analytical methods. Section 4 reports and interprets the emergent dimensions of Food Analytics. Section 5 synthesizes the findings and identifies research opportunities. Section 6 discusses the limitations, and Section 7 concludes.

2. Literature review

2.1 The emergence of food analytics

The emergence of Food Analytics has been driven by the digitalization of food-system activities and the increasing availability of data across production, processing, logistics, retail, consumer interactions, nutrition, and scientific research. Advances in AI, Machine Learning (ML), big-data infrastructures, cloud computing, the Internet of Things (IoT), and decision-support systems have progressively extended analytical activity beyond isolated laboratory and operational applications. This development has created the conditions for a broader domain concerned with monitoring, prediction, optimization, and decision-making across interconnected food-system contexts (Gao *et al.*, 2025; Kohli *et al.*, 2025; Siddique *et al.*, 2025).

Although analytical approaches have long been used in food science, earlier work largely focused on specialized applications such as chemometrics, quality control, food composition analysis, and process optimization; more recently, analytical methods have expanded beyond laboratory settings to address broader challenges in food safety, supply chain management, consumer behavior, sustainability, nutrition, and digital food ecosystems (Kohli *et al.*, 2025; Zhou *et al.*, 2025). This expansion suggests that Food Analytics is evolving from a collection of isolated analytical applications into a broader socio-technical field integrating food science, data science, operations management, nutrition, and information systems.

2.2 Existing reviews of food analytics-related research

The rapid growth of Food Analytics has stimulated a growing body of review literature. Existing reviews have examined the application of AI and ML in food science and industry, highlighting advances in predictive modeling, food quality assessment, process optimization, and intelligent decision support (Gao *et al.*, 2025; Siddique *et al.*, 2025). Other reviews have focused on big data and analytics applications within food supply chains, emphasizing traceability, operational efficiency, sustainability, and data-driven management of food-system activities (Kamble *et al.*, 2020; Kohli *et al.*, 2025).

Additional studies have explored the role of emerging digital technologies within smart food systems. These reviews have investigated the integration of AI, IoT, blockchain, cloud computing, and cyber-physical systems to support transparency, automation, interoperability, and real-time monitoring throughout the food value chain (Soundara Rajan and Wani, 2025; Xhani *et al.*, 2025). Related work has examined sustainability-oriented applications, including smart packaging, intelligent monitoring systems, food waste reduction, and environmentally sustainable food-system management (Gupta *et al.*, 2025; Zhou *et al.*, 2025).

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Consumer- and nutrition-oriented reviews represent another important stream of research. Previous studies have examined consumer analytics, food recommendation systems, sentiment analysis, digital consumer engagement, and personalized nutrition applications supported by ML and data-driven technologies (Kechri *et al.*, 2025; Shaeeali *et al.*, 2020; Stirling *et al.*, 2021). Collectively, these reviews demonstrate the growing influence of analytics across multiple food-system domains and confirm the increasing convergence between food science and digital technologies.

2.3 Limitations of existing knowledge

Despite these contributions, existing knowledge remains fragmented across technologies and application contexts. Reviews generally examine AI, big data, blockchain, IoT, sustainability, supply chains, packaging, consumer behavior, or nutrition independently (Gao *et al.*, 2025; Kechri *et al.*, 2025; Kohli *et al.*, 2025; Siddique *et al.*, 2025). Recent systematic and bibliometric reviews similarly remain concentrated on specific technologies or application domains rather than mapping the emerging structure of Food Analytics as an integrated field. Although this specialization provides detailed insights, it offers only a limited understanding of how these research streams interact or collectively contribute to the thematic structure of Food Analytics.

Most reviews provide a snapshot of the current state of knowledge without explicitly examining emergence. They describe technological developments and application domains but seldom investigate how new themes arise, how established themes transform, or how previously separate areas converge to create new research trajectories. This limitation matters because Food Analytics appears to be an emerging domain whose conceptual boundaries and research priorities are still evolving.

Finally, the multidisciplinary nature of Food Analytics creates conceptual ambiguity. Related concepts such as food informatics, food data science, food intelligence, smart food systems, digital food ecosystems, and analytics-enabled food management are often used interchangeably or with overlapping meanings (Kohli *et al.*, 2025; Siddique *et al.*, 2025; Xhani *et al.*, 2025). Consequently, it remains unclear how these concepts contribute to the formation of a coherent research domain and which themes are shaping its emerging frontier.

To clarify the present study's positioning, Table 1 compares previous reviews in related areas by scope, database, period of analysis, bibliometric technique, approach to emergence, and remaining research gap. The comparison highlights the methodological distinction between mapping established research structures and identifying emerging topics at the descriptor level before thematic clustering. It also establishes the substantive basis for comparison with the present findings: earlier reviews consistently identify advances in food quality and processing, supply-chain and smart-system analytics, consumer and nutrition applications, and digital technologies, but they generally examine these developments within separate technological or application-specific streams. This distinction provides the basis for assessing whether the emergent dimensions identified here reproduce established trajectories, reveal changes within them, or extend them through previously underexamined forms of convergence.

2.4 Research gap and study contribution

Accordingly, the principal unresolved issue is not whether analytical technologies are being adopted, but how recent developments combine to form the emerging research frontier of Food Analytics. Existing reviews have not systematically identified descriptors exhibiting emergence, examined how they combine into thematic clusters, or interpreted the higher-order dimensions produced by their convergence. Relative to previous bibliometric and technology-focused reviews that have examined AI, big data analytics, smart food systems, food supply chains, personalized nutrition, and digital transformation largely as distinct technological or application streams, the present approach advances the literature by

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Table 1. Comparison of previous reviews and positioning of the present study

Study	Review scope	Database	Period	Bibliometric technique	Emergence	Unresolved gap
Kamble *et al.* (2020)	Data-driven sustainable agri-food supply chains	WoS	2000–2017	Descriptive analysis; content classification	None	No field-level emergence analysis
Shaeeali *et al.* (2020)	Food-delivery customer-review analytics	Google Scholar; ScienceDirect; Springer; IEEE	2005–2020	Descriptive classification	None	Application-specific; no thematic emergence
Stirling *et al.* (2021)	Social-media analytics of dietary behavior	Medline; CINAHL; Scopus; ACM; Engineering Village	2014–2020	Descriptive mapping	None	Nutrition-specific; no formal emergence detection
Purwati *et al.* (2025)	Computer vision for nutrition assessment	Scopus	2020–2024	Bibliometric trends; collaboration networks	Topic-frequency trends	Technology-specific; no formal emergence metric
Siddique *et al.* (2025)	Big-data analytics and ML in the food industry	Scopus; WoS; IEEE Xplore; ScienceDirect; Google Scholar	2010–2024	Network and word-cloud analysis	Interpretive identification of emerging themes	No quantitative descriptor-level emergence
Present study	Food Analytics research frontier	Scopus	2016–2026	Co-word analysis; Louvain clustering	Porter descriptor-level emergence	Integrates emergent topics into field-level dimensions

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examining their convergence within a common Food Analytics domain and by identifying the emerging structure that connects them. The study therefore makes an integrated methodological and conceptual contribution, supported by empirical findings. Methodologically, it combines descriptor enrichment and normalization with descriptor-level emergence detection before co-word network construction and community detection, thereby identifying the thematic structure specifically among emergent rather than established research topics. This distinction provides information that conventional bibliometric trend mapping does not directly capture. Frequency, citation, or co-occurrence patterns primarily reveal the prominence and structure of topics already represented in the literature.

In contrast, descriptor-level emergence evaluates whether a concept exhibits a temporally distinctive combination of novelty, persistence, and recent growth. It can therefore distinguish rapidly developing but still comparatively small research signals from highly visible established topics and identify acceleration, recombination, or convergence among concepts before these developments necessarily become dominant in the broader thematic structure. Conceptually, it provides an integrative representation of the developing structure of Food Analytics through four interacting higher-order dimensions. Empirically, the 111 emergent descriptors and five Louvain communities identify the research themes and trajectories from which this conceptual synthesis is derived. Addressing this gap provides early evidence of the research trajectories through which Food Analytics is expanding beyond its established applications and clarifies the field's developing structure.

3. Materials and methods

3.1 Study design

Table 2 summarizes the principal parameters of the study following Kitchenham's (2004) principles of rigor, transparency, and reproducibility. The study is designed as a technological-intelligence investigation using bibliometric tech-mining methods and scoping-oriented retrieval to examine the emergence of Food Analytics research (Aria and Cuccurullo, 2017; Donthu *et al.*, 2021). Bibliographic records are treated as structured science-and-technology information, while normalized descriptors and their temporal and co-occurrence patterns constitute the principal analytical units for identifying emerging R&D signals. Descriptor normalization, emergence detection, co-word analysis, and Louvain clustering are employed to identify and interpret emergent research themes. The objective is to map the conceptual structure of emerging Food Analytics research rather than to evaluate the effectiveness of specific analytical methods, technologies, or interventions. This technological-intelligence design is particularly appropriate for identifying an emerging research frontier because it evaluates temporal emergence at the descriptor level before constructing the thematic network, rather than inferring emergence from the prominence of topics in the overall literature. Consequently, unlike conventional frequency-, citation-, or co-occurrence-based mapping, the approach can identify comparatively small but persistent and rapidly developing research signals before examining how those emergent concepts are relationally organized.

Where applicable, established reporting recommendations were consulted selectively to improve methodological transparency: PRISMA-S informed search-strategy reporting (Rethlefsen *et al.*, 2021) while PRISMA 2020 conventions informed reporting of the identification, screening, eligibility, and inclusion flow (Page *et al.*, 2021). These reporting conventions do not define the study design or imply an intervention-review methodology.

3.1.1 Database selection. Scopus was selected as the sole bibliographic source because of its extensive coverage of research in food science, agriculture, nutrition, computer science, engineering, business, and management, providing multidisciplinary coverage consistent with the heterogeneous disciplinary structure of Food Analytics. Independent database-comparison studies have also documented Scopus's broad multidisciplinary coverage relative to other major bibliographic databases (Mongeon and Paul-Hus, 2016), as well as its widespread use in bibliometric and scientometric studies. This breadth is particularly appropriate for the present

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Table 2. Parameters of the study

Parameter	Value
Database	Scopus
Years of analysis	2016 up to May 2, 2026
Data retrieval	May 2, 2026
Search string	Seven complementary Scopus Boolean queries developed through an iterative, evidence-driven process; complete executable queries, field codes, operators, execution conditions, and final retrieval date are reported in Supplementary File S1
Inclusion criteria	Records were included when: (1) they addressed food, food products, food systems, food supply chains, diets, nutrition, food consumption, food safety, food quality, food production, food services, or agri-food contexts; (2) they explicitly applied, developed, evaluated, or discussed analytical or computational methods such as data analytics, data science, AI, ML, statistical modeling, predictive analytics, data mining, computer vision, natural language processing, recommender systems, optimization, simulation, or decision support; (3) the analytical method was directly linked to a food-related objective; (4) they presented an empirical, methodological, review-based, computational, or decision-support contribution; and (5) Food Analytics was a central rather than incidental focus
Exclusion criteria	Records were excluded when: (1) they did not address a food-related domain; (2) they lacked a substantive analytics or computational component; (3) they were limited to laboratory, chemical, microbiological, sensory, or instrumental analysis without a data-driven modeling component; (4) they focused on general food science without a substantive analytics contribution; (5) analytical methods were applied outside food contexts; (6) they addressed agriculture only, without a direct food-related connection; (7) they concerned clinical nutrition or biomedical outcomes without a Food Analytics contribution; (8) they focused on business, management, sustainability, or supply chains without analytical modeling or decision support; (9) analytical terminology referred only to abstract methodological problems without a substantive food application; or (10) food-related terms appeared only incidentally

objective because the study seeks to identify emerging relationships across these disciplinary boundaries rather than to characterize a single specialized literature. Using a single database was particularly important because the present analysis draws a study-specific subset from a larger bibliographic corpus constructed for the broader investigation of Food Analytics. Maintaining a common source preserves consistent document identifiers, bibliographic fields, descriptors, citation metadata, and indexing conventions across related analyses. Triangulation with Web of Science or OpenAlex would require record deduplication, field harmonization, reconciliation of differences in keyword and indexing structures, and normalization of heterogeneous metadata before descriptor-level analysis. These additional transformations could introduce inconsistencies into descriptor extraction and normalization and complicate comparability across analyses. For this reason, database consistency is prioritized over marginal coverage gains from database aggregation. This choice reduces inconsistencies associated with merging heterogeneous metadata structures and facilitates descriptor extraction, normalization, and thematic analysis. However, reliance on Scopus may exclude relevant research indexed exclusively in other databases, particularly regional or differently indexed publications; this limitation is discussed further in Section 6.

3.1.2 Search string and query construction. The search strategy followed an iterative evidence-driven approach and imposed no publication-year restrictions. The initial query was constructed around ten highly specific expressions directly associated with Food Analytics: “food analytics,” “food data analytics,” “food informatics,” “food data science,” “agri-food analytics,” “agrifood analytics,” “agri-food data analytics,” “agrifood data analytics,” “agri-food informatics,” and “agrifood informatics.” Retrieved records were then examined for their titles, abstracts, author keywords, index keywords, and cited-reference titles to identify additional relevant expressions and refine the query’s Boolean

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structure. Newly identified terms were incorporated, and the search was repeated until successive iterations produced no substantively new concepts or applications relevant to the domain. For transparent reporting of the resulting search framework, terminology represented across the final queries was organized using a PCC-informed structure: Population/source (food-system literature), Concept (analytics, informatics, artificial intelligence, data science, optimization, and decision support), and Context (production, processing, logistics, retail, nutrition, consumer behavior, food safety, and related food-system applications). Table 3 summarizes representative terms from the final search strings within these components. This procedure balanced specificity and coverage while generating a set of complementary search queries that capture the evolving Food Analytics literature. Six queries underwent empirical refinement through repeated execution and inspection of the retrieved records, while a seventh strict query retained only the ten initial high-specificity expressions as a precision-oriented anchor. Table 4 summarizes the principal refinement trajectory of each search, including the source of candidate modifications, inclusion decisions, rationale, retrieval-count changes, and stopping criterion.

Candidate terms and structural modifications were treated as proposals until their effect on retrieval and thematic relevance had been evaluated. Researchers made final inclusion or exclusion decisions based on inspection of the resulting records, alignment with the operational scope of Food Analytics, and the balance between coverage and off-domain retrieval. When alternative formulations produced conflicting indications regarding coverage and precision, the search was re-executed with the competing formulation, and the retrieved corpus was inspected before a final decision was made. Query development stopped when further refinement produced no substantively new domain-relevant terminology or when additional restriction would compromise complementary coverage. Complementary queries also reduced dependence on any single label for Food Analytics by capturing alternative terminology and adjacent application vocabularies. Such terminology does not necessarily denote equivalent research domains: labels such as food informatics, food intelligence, and smart food systems may overlap with Food Analytics while also encompassing research outside its operational scope. Including them in query development could increase coverage while also increasing off-domain retrieval, making empirical inspection and eligibility screening necessary for corpus delimitation. Nevertheless, retrieval remained terminology-dependent, and relevant studies expressed exclusively through terms not represented in the final queries may have been omitted.

Table 3. PCC-informed conceptual organization of the search terminology

PCC component	Scope	Representative terms from final queries
Population/ source	Food-system literature	Food; food system(s); food industry; food supply chain(s); agri-food; agrifood; agro-food; diet; meal; recipe
Concept	Analytics and computational methods	Analytics; data analytics; data science; informatics; computing; big data; AI; machine learning; deep learning; computer vision; predictive analytics; data mining; neural networks; chemometrics; predictive modeling; multivariate analysis; NLP; text mining; sentiment analysis; knowledge graphs; recommender systems; optimization-oriented forecasting
Context	Food-system applications	Food safety; food quality; food security; processing; consumption; retail; markets; supply chains; nutrition; dietary analysis; restaurants; grocery; demand forecasting; food waste/ loss; authenticity; adulteration; shelf life; freshness; sensory/ flavor analysis; personalized recommendation

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Table 4. Summary of iterative development of the food analytics search strategy

Search	Candidate modification/source	Decision and rationale	Retrieval trajectory	Stopping criterion
Strict	Ten explicit Food Analytics expressions	Retained unchanged as a high-precision anchor	58	No iterative expansion intended
Main 1	Terms and proximity structures refined from retrieved titles, abstracts, and keywords	Broad AI/ML formulations rejected as noisy; food–analytics proximity retained	58 → >106,000 → 1,592 → 810 → 702 → 702	Retrieval stabilized after further phrase expansion produced no increase
Main 2	Field and proximity formulations tested against retrieved-record terminology	Broad abstract intersections rejected; TITLE/KEY proximity retained to balance precision and recall	58 → 2,634 → 244 → >64,000 → >24,000 → 1,563	Precision–coverage convergence after corpus inspection
Main 3	Candidate expressions evaluated through corpus inspection and domain relevance	Off-domain nutrition, image, generic agri-food, and other noisy expressions removed; food-analytics-specific terms retained	2,067 → 2,945 → 446 → 223 → 4,684 → 228	Successive inspections yielded no new domain-relevant terms
Complementary 1	Gaps identified by comparing the main searches and inspecting uncovered application areas	Broad prediction terms reduced; recommendation expressions restricted to improve precision	721 → 764 → 540	Further restriction risked excluding targeted missing subdomains
Complementary 2	Missing property-level analytics identified through comparative gap analysis	Broad omics/vision formulation rejected; sensory, compositional, authenticity, and chemometric terms retained	5,735 → 130	Retrieved corpus concentrated on the intended property-level analytics
Complementary 3	Missing methodological/ application areas identified through comparison of main queries	Broad intersections progressively constrained with domain-specific anchors	2 → 7,952 → 789 → 19,270 → 12,753 → 9,168 → 320	Successive corpus inspection yielded no new relevant search concepts

3.2 Data collection and preliminary preparation

The unrestricted Scopus search retrieved 2,913 records. During screening, 263 records were excluded based on metadata criteria, including publication year and document type, leaving 2,650 documents for eligibility assessment, as summarized in the PRISMA flowchart in Figure 1. Titles and abstracts were subsequently evaluated against the eligibility criteria summarized in Table 2. Records were retained when data-driven, computational, statistical, AI, ML, optimization, or decision-support approaches were applied substantively in food-system contexts. Records were excluded when the analytical terminology primarily referred to conventional laboratory measurements, generic agricultural technologies without a food-analytics focus, or computational applications unrelated to food systems.

The temporal restriction to publications from 2016 onward was applied during metadata screening because the study examines the recent emergence of themes. The

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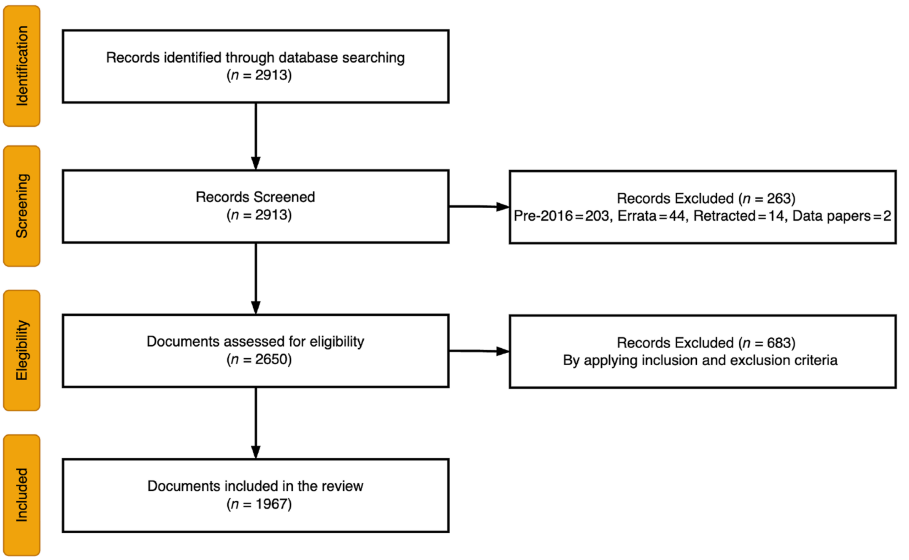


Figure 1. The PRISMA flow chart

subsequent eligibility assessment excluded 683 additional records, yielding a final analytical corpus of 1,967 publications published between 2016 and May 2, 2026. Figure 1 documents the complete identification, screening, eligibility, and inclusion process.

A single researcher conducted screening in two stages with computational support. The first stage applied computational and metadata-based filtering to identify records meeting predefined exclusion conditions, followed by researcher validation of the resulting exclusions. The second stage assessed titles and abstracts against the operational definition of Food Analytics and the predefined eligibility criteria. Borderline decisions were resolved by determining whether the analytical or computational component made a substantive contribution to a food-system objective; for example, agriculture-focused records without a direct food-related connection and clinical-nutrition records without a Food Analytics contribution were excluded, whereas records were retained when the analytical method was substantively linked to a food-related objective. Computational filtering supported identifying candidate exclusions, while researchers made final eligibility decisions. Screening quality control therefore relied on the consistent application of predefined eligibility criteria across records, computational identification of candidate exclusions followed by researcher validation, and a separate title-and-abstract assessment against the operational definition of Food Analytics, thereby providing a structured decision process to minimize subjective variation. Because multiple reviewers did not conduct screening independently, inter-reviewer consensus procedures and agreement statistics were not applicable.

Metadata from the seven Scopus searches were consolidated by removing exact duplicate records across the combined search outputs. All Scopus document types were retained except those explicitly excluded in the PRISMA flowchart. Retraction status was determined from the corresponding status information provided in the Scopus CSV export. Early-access and articles-in-press records were treated as standard bibliographic records and retained because they provide relevant recent evidence for identifying emerging research signals. The original Scopus export files were preserved unchanged as a raw-data snapshot before metadata

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filtering, eligibility assessment, descriptor normalization, and subsequent analytical transformations, enabling the processed corpus to be traced back to the originally retrieved records.

3.3 Descriptor preparation

All bibliographic records were exported from Scopus in CSV format, including titles, abstracts, author keywords, index keywords, affiliations, references, and publication years. Author and index keywords were combined with noun phrases extracted from titles and abstracts to create a unified descriptor field. Noun-phrase extraction was performed through a multi-stage natural-language-processing pipeline integrating complementary tokenization, linguistic parsing, candidate phrase extraction, and named-entity recognition procedures. Because extraction relies on multiple complementary procedures rather than a single parser configuration, it is not characterized by one tokenizer, one fixed noun-phrase pattern, or a global minimum/maximum token-length rule. No pre-normalization frequency threshold was applied. Multi-word expressions were preserved because they provide greater conceptual specificity than isolated terms.

Descriptor preparation combined automated processing with manual validation. A controlled thesaurus was developed to standardize spelling variants, punctuation, spacing, capitalization, singular and plural forms, hyphenation variants, acronyms, and conceptual synonyms (Courseault Trumbach and Payne, 2007; Porter *et al.*, 2019; Zhang *et al.*, 2014). The normalization process combined automatic and manual stages. Because normalized descriptors constitute the analytical units on which emergence and co-occurrence are subsequently calculated, descriptor preparation was treated as a conceptual measurement stage rather than as a purely lexical cleaning operation. First, deterministic rules automatically merged descriptors differing only in features such as uppercase/lowercase usage, punctuation, spacing, hyphenation, and singular/plural forms. Next, similarity-based procedures, including fuzzy matching and stem variations, identified candidate matches among descriptors. Stemming or lemmatization was not used as a general replacement for descriptor normalization; stem-based similarity was used only to identify candidate matches for subsequent validation. These candidate matches were not merged automatically; lexical similarity served only to generate candidates for validation. Each candidate was manually reviewed, and consolidation required conceptual equivalence in the bibliographic context before inclusion as a many-to-one mapping in the thesaurus; lexically similar descriptors representing distinct concepts were retained separately. Acronyms and conceptual synonyms were treated similarly. For example, “AI” and “artificial intelligence,” “agri-food” and “agrifood,” and singular/plural variants were consolidated under common descriptors. The thesaurus was iteratively refined as additional equivalent expressions were identified during descriptor and cluster inspection. Generic and cross-cutting expressions that lacked discriminative thematic value were removed as stopwords. The descriptor-construction stage initially generated 47,795 distinct raw concepts; after the complete cleaning and normalization workflow, the analytical vocabulary contained 35,493 normalized concepts. These figures characterize the descriptor space before and after the complete workflow and should not be interpreted as counts of thesaurus substitutions, synonym merges, stopword removals, or other mutually exclusive transformation categories. The resulting normalized descriptors served as the analytical units for emergence detection and thematic clustering.

3.4 Quality appraisal

Consistent with the study’s mapping-oriented design, no formal quality, risk-of-bias, or effectiveness assessment was conducted; the findings indicate thematic emergence rather than the superiority of particular technologies or interventions.

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3.5 Data analysis and interpretation

3.5.1 Identification of emergent descriptors. This research uses the emergence indicators proposed by Porter *et al.* (2019). The analysis was conducted on normalized descriptors extracted from the complete corpus covering 2016 to May 2, 2026. Following Porter *et al.* (2019), we treated emergence as a temporal property of descriptor activity rather than simple recent frequency. In this application, novelty captures the extent to which a descriptor is comparatively weakly represented in the baseline period relative to its more recent activity; growth captures the increase in descriptor activity between the baseline and recent periods; and persistence requires that the descriptor occur across multiple annual periods rather than appear as an isolated short-lived signal. A descriptor was therefore classified as emergent only when these temporal conditions were jointly satisfied together with a minimum level of overall activity. The final configuration used a three-year baseline period, a four-year recent period, a novelty threshold of 0.30, a minimum total frequency of seven records, activity in at least three annual periods, and a recent-to-baseline growth ratio threshold of 1.12. Operationally, the novelty threshold constrained prior activity, the minimum of three active annual periods implemented the persistence requirement, the minimum total frequency excluded signals with insufficient empirical support, and the recent-to-baseline ratio required recent activity to exceed baseline activity by the specified growth criterion. The configuration therefore combined the conceptual emergence criteria derived from Porter *et al.*'s (2019) framework with study-specific operational thresholds intended to exclude sparse or short-lived signals; these fixed criteria were not selected through multidimensional empirical optimization. The growth ratio threshold was selected through sensitivity analysis across values ranging from 1.00 to 1.20 in increments of 0.01; 1.12 provided the best balance between interpretability and consistency of the resulting thematic clusters. These parameters enabled the identification of descriptors that exhibited sustained recent growth while maintaining sufficient historical continuity to represent meaningful research developments rather than isolated observations.

3.5.2 Emergent theme identification. Descriptors classified as emergent were used to construct a weighted, undirected co-occurrence network at the document level. Nodes represented emergent descriptors, while edge weights corresponded to the number of documents in which descriptor pairs co-occurred. Louvain community detection was applied to identify groups of closely related emergent descriptors, producing the emergent thematic clusters examined in this study.

Sensitivity analysis confirmed that the selected emergence threshold produced conceptually coherent clusters that remained stable across neighboring parameter values.

3.5.3 Interpretation of emergent themes. Emergent clusters were initially interpreted using their most frequent descriptors, internal co-occurrence relationships, and the associated document content. For each higher-order analytical dimension, the corresponding emergent descriptors were used to retrieve the associated records, which were ordered by publication date, and the 100 most recent abstracts were examined to identify recurring applications, methodological relationships, research directions, and emerging ideas. Because several communities overlapped conceptually, their shared applications and methodological relationships were compared and consolidated into four higher-order analytical dimensions. The dimensions reported in Section 4 are therefore interpretive syntheses of the Louvain communities rather than direct clustering outputs.

4. Results and discussion

Louvain community detection identified five emergent thematic clusters. These clusters comprise 111 emergent descriptors, all of which satisfied the novelty, persistence, and recent-growth criteria defined in Section 3.5.1 using Porter *et al.*'s (2019) emergence framework. Four clusters corresponded predominantly to the substantive research trajectories interpreted as intelligent food quality and composition analytics; predictive and operational food-system analytics; personalized nutrition and consumer analytics; and semantic, generative, and

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knowledge-driven food analytics. The fifth cluster was primarily methodological, encompassing ML, Deep Learning (DL), explainable AI, optimization, multimodal data integration, model validation, and analytical infrastructure. Because these methods supported applications across all four substantive trajectories, the cluster was not retained as an independent dimension. Still, it was integrated into the dimensions in which its methods were applied. The resulting four-dimensional framework therefore preserves the main substantive structure of the five-cluster solution while accounting for the expected thematic overlap produced by shared analytical methods. Table 5 provides quantitative support for this interpretation by reporting the number of emergent descriptors in each cluster (33, 27, 24, 16, and 11, respectively), together with representative descriptors that characterize their thematic content.

At a higher conceptual level, the four dimensions represent complementary analytical functions within the developing socio-technical structure of Food Analytics. Intelligent food quality and composition analytics provides an evidence-generating function by transforming physical, chemical, biological, and compositional properties into analytically usable information. Predictive and operational food-system analytics extend this evidence toward system-level forecasting, coordination, optimization, and decision support. Personalized nutrition and consumer analytics introduce a human-centered function by incorporating individual characteristics, preferences, behaviors, and contexts into food-related analytical decisions. Semantic, generative, and knowledge-driven food analytics provide a knowledge-integration function that represents, connects, interprets, and reuses heterogeneous evidence across applications. Together, these dimensions characterize the evolution of Food Analytics not simply as the adoption of increasingly sophisticated technologies, but as the growing integration of food-system evidence, analytical intelligence, decision processes, human contexts, and knowledge infrastructures within a socio-technical research field. This structure reflects the convergence of food science, information systems, artificial intelligence, consumer analytics, and decision sciences, each contributing distinct but interdependent forms of domain knowledge, analytical capability, infrastructure, or decision support. AI therefore operates as a transversal analytical capability across the dimensions rather than defining Food Analytics as merely an application domain of AI. The dimensions are also functionally interdependent rather than isolated thematic domains. Evidence generated through quality and composition analytics can support predictive, operational, and personalized applications; consumer and operational data provide additional context for analytical decisions; and semantic and generative architectures can connect and reuse

Table 5. Quantitative characterization of emergent thematic clusters

Cluster	Number of descriptors	Representative emergent descriptors
1	33	artificial intelligence; food supply chain; food waste; Internet of Things; food security; food processing; food traceability; blockchain; food quality and safety; climate change
2	27	machine learning; learning systems; machine learning models; explainable artificial intelligence; extreme gradient boosting; data science; food fraud; supervised machine learning; attention mechanisms; Shapley additive explanations
3	24	personalized nutrition; eating habits; object detection; multi-modal; nutritional analysis; dietary preferences; health and nutrition; chatbot; individual preferences; federated learning
4	16	agricultural practices; chemistry; functional food; nutritional science; pesticides; hyperspectral images; bioactive compounds; pesticide residues; peptides; therapy
5	11	language models; dietary planning; artificial intelligence models; graph neural networks; generative artificial intelligence; process control; recipe data; ChatGPT; lipid; user engagement; heterogeneous graph

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knowledge produced across these activities. These interactions suggest that Food Analytics is evolving from relatively specialized analytical applications toward a more interconnected analytical ecosystem in which evidence, prediction, decision support, personalization, and knowledge integration increasingly reinforce one another.

4.1 Dimension 1: intelligent food quality and composition analytics

Intelligent food quality and composition analytics constitute a major research dimension in recent developments in Food Analytics. It encompasses the application of AI, ML, advanced sensing technologies, and analytical chemistry to evaluate food composition, quality, authenticity, safety, and functional properties. This dimension reflects a transition from conventional laboratory-based and single-modality analyses toward integrated, data-intensive systems capable of processing complex chemical, spectral, biological, and sensor-generated information and linking these data to faster, non-destructive, and more analytically comprehensive assessments of food integrity and functionality (Aboyeji *et al.*, 2026; Mehdizadeh *et al.*, 2026; Xiao *et al.*, 2026; Zhang, 2026).

A defining feature of this dimension is the convergence of computational intelligence with advanced analytical instrumentation. DL, hyperspectral imaging, electrochemical sensing, spectroscopy, chemometrics, and multimodal data integration are increasingly combined to support rapid, non-destructive, and high-throughput assessment of foods and ingredients (Ahmad *et al.*, 2026; Xue *et al.*, 2026; Zhang, 2026). These techniques extend traditional food testing by detecting subtle patterns in high-dimensional datasets that may be difficult to identify with conventional statistical or laboratory methods.

Principal applications include food classification, compositional and nutritional analysis, contaminant detection, adulteration identification, authenticity verification, and quality evaluation (Ayranci and Erkmen, 2026; Shanmugapriya *et al.*, 2026; Shin *et al.*, 2026). Predictive models associate spectral, chemical, sensory, and microbiological signals with food attributes, enabling faster assessment and supporting product classification, fraud detection, origin verification, and quality assurance across production and distribution (Sukmawani *et al.*, 2026; Yao *et al.*, 2026; Zhu *et al.*, 2026). These applications connect laboratory measurement with broader monitoring processes while retaining an analytical focus on food integrity and composition.

The dimension also incorporates research on ingredients, bioactive compounds, and functional foods. Computational and analytical methods characterize peptides, phytochemicals, lipids, nutrients, and other compounds associated with technological or health-related functionality (Alzunaidy, 2026; Ekrami *et al.*, 2025; Gunapati *et al.*, 2026; Yu *et al.*, 2025). This expands food composition analysis beyond proximate measurement towards the identification of complex molecular profiles and potential biological functions. The increasing incorporation of multi-omics, spectral, and physico-chemical data illustrates the gradual integration of food chemistry, informatics, and biological analysis within a common analytical space (Ahmad *et al.*, 2026; Zhang, 2026).

Despite methodological progress, this dimension remains constrained by many models' dependence on small, homogeneous, or product-specific datasets, which limits transfer across foods, instruments, and production environments (Mehdizadeh *et al.*, 2026; Shanmugapriya *et al.*, 2026; Zhang, 2026). Differences in sample preparation, measurement protocols, sensor configurations, and preprocessing procedures further reduce reproducibility and complicate calibration transfer. These limitations are especially consequential in hyperspectral, electrochemical, and other sensor-based applications, where instrument variability and environmental conditions can materially alter model performance.

4.2 Dimension 2: predictive and operational food-system analytics

Predictive and operational food-system analytics apply Food Analytics to planning, monitoring, coordination, and optimization across agricultural production, post-harvest

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handling, logistics, inventory, distribution, and market-facing activities. It reflects a transition from reactive management towards adaptive systems that combine AI, ML, IoT, blockchain, sensors, enterprise data, and digital platforms to forecast demand, identify inefficiencies, anticipate operational outcomes, and support real-time decision-making (Borugadda and Kalluri, 2026; Budhathoki *et al.*, 2026; Zhao *et al.*, 2026). Unlike quality and composition analytics, it primarily analyzes the performance of interconnected processes rather than the attributes of individual food products.

Supply-chain analytics combines predictive optimization with traceability and information visibility. ML and optimization models support forecasting, inventory management, logistics coordination, distribution planning, and resource allocation, enabling organizations to anticipate disruptions and improve product flows (Braggio *et al.*, 2026; Kanicheril Ambikalekshmi *et al.*, 2026; Nguyen *et al.*, 2026; Zhu *et al.*, 2026). In parallel, IoT infrastructures, blockchain, and integrated platforms record transactions and track products across organizational boundaries, improving transparency and coordination across food networks (Brar *et al.*, 2026; Mia *et al.*, 2026; Zhang, 2026). Analytics therefore supports both operational prediction and the information infrastructure required to manage interconnected supply chains.

Agricultural and post-harvest operations also form an important component of this dimension. Research applies AI-enabled monitoring and forecasting techniques to improve agricultural productivity, resource management, harvest planning, and post-harvest handling (Ahmad *et al.*, 2026; Borugadda and Kalluri, 2026; Zhang, 2026). Sensor networks, remote monitoring technologies, and data-driven decision-support tools are increasingly used to support operational decision-making under uncertain and variable environmental conditions. These developments extend predictive analytics beyond individual facilities to broader production systems involving multiple stakeholders and interconnected operational processes.

Methodologically, the dimension is characterized by analytical models designed for forecasting, optimization, anomaly detection, and real-time operational decision support. These models integrate sensor streams, transactional records, logistics data, and environmental information to anticipate disruptions, coordinate resources, and improve planning across interconnected food-system activities (Ablegue *et al.*, 2026; Ahmad *et al.*, 2026; Kanicheril Ambikalekshmi *et al.*, 2026). Their practical value depends not only on predictive accuracy but also on the interpretability and actionability of their outputs within organizational decision processes.

Despite their rapid expansion, predictive and operational food-system analytics face several limitations. Operational datasets are frequently fragmented across organizations, technologies, and geographic regions, complicating data integration and interoperability (Lu *et al.*, 2026; Wani *et al.*, 2026; Zhou *et al.*, 2026). Variations in data quality, annotation practices, and technological infrastructures further complicate the development of transferable analytical solutions. In addition, the complexity of food systems creates challenges associated with model generalizability, explainability, and implementation across heterogeneous operational environments (Li *et al.*, 2026; Uzunov and Angeleska, 2026; Zhao *et al.*, 2026). Regulatory differences, infrastructure constraints, and the distributed nature of food networks add further complexity to large-scale deployment.

4.3 Dimension 3: personalized nutrition and consumer analytics

Personalized nutrition and consumer analytics apply Food Analytics to understanding dietary behavior, nutritional needs, food choices, and consumer preferences at the individual level. Unlike operational food-system analytics, which focuses on managing processes and supply chains, this dimension centers on consumers as the primary unit of analysis. The dimension reflects the convergence of AI, ML, multimodal data integration, and digital health technologies to provide individualized insights and recommendations for diet, nutrition, and food consumption (Singh *et al.*, 2026; Srivastava *et al.*, 2026; Wang *et al.*, 2026).

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A defining characteristic of this dimension is the shift from population-level dietary guidance to adaptive, personalized nutritional support. Recent research increasingly employs computational models that integrate diverse sources of information, including food images, electronic health records, dietary logs, textual descriptions, physiological indicators, sensory information, and user profiles (Bhuiyan *et al.*, 2025; Dakhia *et al.*, 2025; Ekrami *et al.*, 2025; Jansen and Bennin, 2025). By combining these heterogeneous data sources, analytical systems can generate individualized assessments of dietary intake, nutritional status, and food-related behaviors that complement traditional survey-based approaches with more individualized evidence.

Food recognition and dietary assessment constitute important research streams within this dimension. Advances in computer vision, DL, and image analytics have enabled automated food identification, portion estimation, meal classification, and nutritional assessment using photographs captured through mobile devices and digital platforms (Gaonkar and Sangeetha, 2026; Jagadeesh *et al.*, 2026; Ju *et al.*, 2026). These approaches aim to reduce the burden of conventional dietary monitoring while providing more detailed and continuous information about eating behavior. The increasing use of object-based recognition systems illustrates the growing role of AI in transforming food consumption data into actionable nutritional information (Dakhia *et al.*, 2025; Hasdyna *et al.*, 2026; Yin *et al.*, 2025; Zhang *et al.*, 2025).

Personalized recommendation systems combine nutritional, behavioral, contextual, and preference data to suggest foods, meals, recipes, and dietary strategies aligned with individual needs and goals (Bagozi *et al.*, 2026; Fu *et al.*, 2026; Gavai and Heringa, 2026; Lang and Namahoot, 2026). Mobile applications, conversational agents, chatbots, and dietary platforms deliver these recommendations, support behavioral feedback, and facilitate continuous monitoring (Chen and Chiang, 2025; De Croon *et al.*, 2025). Federated learning and privacy-aware architectures further seek to balance adaptive personalization with the protection of sensitive health and behavioral information (Gaonkar and Sangeetha, 2026; Jagadeesh *et al.*, 2026; Sanatbyek *et al.*, 2025).

Consumer analytics also extends beyond nutrition to encompass food preferences, purchasing behavior, and decision-making processes. Analytical models increasingly incorporate behavioral, contextual, and demographic information to examine how consumers evaluate food products and make dietary choices (Qiao *et al.*, 2025; Xiao *et al.*, 2026). This perspective reflects a broader understanding of food consumption as a multidimensional phenomenon influenced by health considerations, personal preferences, environmental factors, and social contexts. Consequently, consumer analytics increasingly examines behavioral patterns and individualized food-related decision-making.

Methodologically, this dimension integrates food recognition, dietary assessment, recommender systems, and behavioral modeling. These approaches combine food images, dietary records, health indicators, preferences, and contextual information to estimate intake, identify consumption patterns, and generate personalized nutritional guidance (Fu *et al.*, 2026; Siripurapu *et al.*, 2026). The growing availability of consumer-generated and interaction data has enabled more adaptive models, while increasing the importance of explainability and privacy when recommendations may influence health-related decisions.

Despite rapid development, several limitations remain. Consumer-related datasets are often fragmented across platforms and applications, complicating data integration and interoperability (Ahmad *et al.*, 2026; Wang and Sheng, 2026; Zhao *et al.*, 2026). Food recognition systems continue to face challenges related to image quality, portion estimation, cultural variability, and food diversity (Dakhia *et al.*, 2025; Purwati *et al.*, 2025; Qiao *et al.*, 2025). In addition, the increasing complexity of personalized recommendation systems raises concerns regarding transparency, interpretability, privacy, and the reliable application of analytical models across diverse populations and contexts (Alkhrijah *et al.*, 2026; Lu *et al.*, 2026; Siripurapu *et al.*, 2026).

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4.4 Dimension 4: semantic, generative, and knowledge-driven food analytics

Semantic, generative, and knowledge-driven food analytics represent the most recent and conceptually distinct dimension emerging within Food Analytics. While earlier dimensions focus on food composition, operational processes, or consumer behavior, this dimension centers on representing, integrating, interpreting, and generating food-related knowledge. This dimension is characterized by the convergence of AI technologies—particularly large language models (LLMs), generative AI, knowledge graphs, ontologies, and graph-based learning systems—with heterogeneous sources of structured and unstructured food data (Gjorgjevikj *et al.*, 2026; Wang *et al.*, 2026).

A defining characteristic of this dimension is the semantic integration of heterogeneous food information rather than prediction or classification alone. Knowledge graphs, ontologies, and graph-based frameworks connect ingredients, foods, nutrients, dietary patterns, processing data, consumer preferences, and health knowledge across structured and unstructured textual, visual, nutritional, behavioral, and process sources (Ding *et al.*, 2026; Ozlu Karahan *et al.*, 2025; Revesai and Kogeda, 2026). These interconnected representations support reasoning, information retrieval, context-aware interpretation, recommendation, and decision support across otherwise disconnected datasets.

The emergence of large language models and generative AI has accelerated the development of this dimension. Recent studies employ conversational agents, generative systems, and language-based analytical frameworks to support food-related information processing, recipe generation, dietary guidance, and interactive decision support (Bagozi *et al.*, 2025; Cheng *et al.*, 2025; Gjorgjevikj *et al.*, 2025). Unlike traditional predictive models that primarily classify or forecast outcomes, generative systems produce new content, explanations, recommendations, and knowledge representations derived from extensive food-related corpora. Consequently, Food Analytics increasingly incorporates technologies that can both analyze and generate information, expanding analytical applications beyond conventional data-processing tasks.

This dimension also connects strongly with decision-support applications across the other three dimensions. Semantic representations can connect information on food composition, processing, supply chains, nutrients, and consumption. Simultaneously, generative interfaces enable users and organizations to retrieve, interpret, and contextualize that knowledge (Bagozi *et al.*, 2025; Gjorgjevikj *et al.*, 2025; Xiao *et al.*, 2026). Rather than constituting another personalized-recommendation stream, this dimension provides knowledge infrastructures that link, query, explain, and reuse heterogeneous food evidence in analytical systems.

Methodologically, this dimension differs from earlier stages of Food Analytics by emphasizing representation learning, knowledge organization, graph modeling, natural language processing, and generative reasoning. While ML and DL remain important, they are increasingly complemented by symbolic representations, semantic modeling approaches, and hybrid analytical architectures that combine statistical and knowledge-based techniques (Gjorgjevikj *et al.*, 2026; Revesai and Kogeda, 2026; Yusufoğlu *et al.*, 2026). This methodological diversification reflects a broader shift towards analytical systems capable of interpreting meaning, context, and relationships rather than merely identifying patterns within data.

Several challenges accompany the development of this dimension. Food-related information is inherently heterogeneous, originating from scientific literature, databases, consumer-generated content, health records, recipes, product descriptions, and operational systems. Integrating these diverse sources into coherent semantic frameworks remains difficult due to differences in terminology, structure, granularity, and data quality (Gjorgjevikj *et al.*, 2026; Xiao *et al.*, 2026). Semantic misalignment, limited domain-specific corpora, and inconsistencies across food vocabularies further complicate the development of interoperable knowledge systems (Balakrishnan *et al.*, 2025; Revesai and Kogeda, 2026; Xiong *et al.*, 2026). In addition, the growing use of generative AI raises concerns about explainability, reliability, bias, privacy, and regulatory compliance, particularly when systems generate

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recommendations or interpretations that directly influence food-related decisions (Ayoub, 2026; Sharma and Gupta, 2025; Zhao *et al.*, 2025).

5. Synthesis and research opportunities

Together, the four dimensions show that the emerging frontier of Food Analytics is structured around an increasingly integrated analytical architecture. Food properties are measured using advanced sensing and computational models; operational data coordinate production and supply chain activities; consumer and health data support individualized analysis; and semantic infrastructures connect these evidence types through knowledge representation and generative interfaces. The field is therefore evolving not simply by adopting additional technologies, but by progressively integrating measurement, prediction, decision support, and knowledge organization across food-system contexts (Ahmad *et al.*, 2026; Zhang, 2026). This trajectory should not be interpreted as a sequential stage model in which one analytical paradigm replaces another. Measurement, prediction, personalization, and knowledge generation coexist and increasingly interact, with later developments building on and recombining capabilities established in earlier analytical traditions. The evidence therefore supports a cumulative and integrative evolution of Food Analytics rather than a deterministic progression through discrete stages.

From a practical perspective, these research directions provide complementary capabilities for decision-making across the food system. For food-industry actors, predictive, multimodal, and real-time analytics can support process control, demand and inventory forecasting, resource allocation, product development, and operational adaptation. In food safety and quality management, AI-enabled sensing, computer vision, spectroscopy, and traceability systems can strengthen contaminant and fraud detection, authenticity assessment, quality monitoring, and preventive risk management. Personalized nutrition services can integrate dietary, behavioral, biometric, and contextual data to provide adaptive food recognition, dietary assessment, recommendation, and health-oriented decision support. Across supply chains, integrating predictive analytics, IoT, blockchain, and interoperable data infrastructures can improve forecasting, inventory management, traceability, transparency, and resilience to disruptions. For technology developers, the empirical frontier emphasizes the need for analytical systems that can integrate heterogeneous food data while remaining explainable, reliable, interoperable, and deployable across different operational environments. For researchers, the intersections among the four dimensions create opportunities to link compositional, operational, consumer, and knowledge-based evidence and to develop infrastructures that enable reuse of analytical knowledge across applications. For policymakers and regulators, explainable and knowledge-driven analytics can strengthen evidence-based food safety oversight, nutritional interventions, traceability requirements, and governance frameworks for responsible and equitable deployment of AI-enabled food technologies.

The four dimensions both confirm and extend previous conceptualizations of digital transformation in the food sector. They confirm the continuing importance of food-quality analysis, operational and supply-chain analytics, consumer and nutrition applications, and digital technologies identified in earlier reviews. However, they challenge interpretations of Food Analytics as a collection of largely independent applications or as the simple transfer of AI technologies into food contexts. Instead, their coexistence and interdependence suggest an emerging interdisciplinary socio-technical domain. From a food-science perspective, intelligent quality and composition analytics provides the measurement and domain-knowledge foundation on which many subsequent analytical decisions depend. From an information-systems perspective, predictive and operational analytics together with semantic and knowledge-driven infrastructures emphasize data integration, interoperability, information flows, and decision support across organizational and application boundaries. Personalized nutrition and consumer analytics introduce a human-centered layer in which

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analytical outputs are adapted to individual characteristics, behaviors, and decision contexts. From a technology-innovation perspective, the four dimensions indicate that advancement increasingly results from recombining sensing, AI, operational data, consumer information, and semantic technologies rather than from the isolated adoption of individual tools. Food Analytics can therefore be conceptualized not merely as an application area of AI, but as a convergent field in which food-science knowledge, information infrastructures, analytical intelligence, and decision processes jointly produce new capabilities across the food system.

This conceptualization also provides a bridge between technological innovation and practical application: the four dimensions identify how advances in sensing and analytical modeling, predictive intelligence, consumer-oriented analytics, and semantic and generative technologies become operational capabilities for measuring food properties, coordinating food-system processes, supporting individualized decisions, and integrating knowledge for decision support. The contribution is therefore not limited to identifying emerging technologies; it explains how their convergence expands the analytical capabilities through which food-system actors can translate heterogeneous data into evidence, predictions, decisions, and reusable knowledge.

At the societal level, Food Analytics can contribute to more secure, healthy, sustainable, and inclusive food systems. Predictive and real-time analytics can strengthen food security by improving resource use, anticipating supply disruptions, reducing losses and waste, and enhancing the resilience of food systems. Consumer health can benefit from more accurate dietary assessment, personalized nutrition, food safety monitoring, and adaptive recommendations that support healthier food choices. Integrating sustainability indicators with operational analytics can also inform resource-efficient production, waste valorization, environmental impact assessment, and circular food-system strategies. However, realizing these benefits equitably requires addressing disparities in digital infrastructure, data availability, technological capabilities, and affordability. Open, interoperable data resources; computationally efficient technologies; inclusive system design; and deployment strategies adapted to resource-constrained settings are therefore important to prevent intelligent food technologies from reinforcing existing inequalities.

The dimensions are functionally interdependent because outputs generated in one area increasingly become inputs for another. Composition and quality data support operational control and nutritional assessment; supply-chain and process data contextualize product performance and availability; consumer and health data refine demand and dietary analysis; and semantic infrastructures enable these heterogeneous sources to be connected and interpreted across applications (Gjorgjevikj *et al.*, 2026; Revesai and Kogeda, 2026; Wei *et al.*, 2026). This interdependence suggests that Food Analytics maturation will depend less on isolated model improvements than on the capacity to coordinate data, analytical outputs, and decisions across scientific, industrial, and consumer settings.

Comparison with previous reviews reveals both continuity and extension. The intelligent food quality and composition dimension aligns with earlier reviews emphasizing AI- and ML-enabled food processing, quality assessment, and process optimization (Gao *et al.*, 2025; Siddique *et al.*, 2025). The predictive and operational food-system dimension similarly extends established work on data-driven supply chains, traceability, sustainability, and smart food systems (Kamble *et al.*, 2020; Kohli *et al.*, 2025; Soundara Rajan and Wani, 2025; Xhani *et al.*, 2025). Personalized nutrition and consumer analytics also build on previously documented trajectories in food recommendation, consumer behavior, social-media analytics, and individualized dietary support (Kechri *et al.*, 2025; Shaeeali *et al.*, 2020; Stirling *et al.*, 2021). The principal difference, therefore, is not that these application areas are entirely new, but that this analysis identifies them as simultaneously emergent and structurally interconnected within a broader Food Analytics domain. The semantic, generative, and knowledge-driven dimension most clearly extends earlier review structures by connecting knowledge graphs, ontologies, large language models, and generative interfaces with quality, operational, nutritional, and consumer data. Taken together, these findings extend

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technology-centered accounts of food digitalization toward a field-level interpretation in which Food Analytics develops through integrating measurement, prediction, personalization, and knowledge representation rather than through the independent advancement of individual technologies.

Several barriers recur across all clusters. These include fragmented, sparsely annotated datasets; weak interoperability; inconsistent measurement and data-curation protocols; limited model transferability; insufficient explainability; and difficulty validating systems across products, regions, populations, and operational environments (Ahmad *et al.*, 2026; Li *et al.*, 2026; Uzunov and Angeleska, 2026). Privacy, algorithmic bias, computational costs, regulatory uncertainty, and unequal access to digital infrastructures further constrain implementation, particularly when analytical outputs affect health, consumer choice, or multi-actor food-system decisions (Ayoub, 2026; Bhuiyan *et al.*, 2025; Zhao *et al.*, 2025). These tensions indicate that technological emergence should not be equated with operational maturity or demonstrated impact. Practical adoption also depends on effective data governance, algorithmic transparency, privacy protection, organizational capabilities, and the technical and institutional conditions required to integrate analytical systems into decision processes. Responsible AI deployment consequently requires attention not only to model performance but also to transparency, accountability, privacy, fairness, human oversight, and differences in organizational capabilities and digital access.

The detected emerging dimensions suggest four interconnected research directions. First, the convergence of intelligent food quality and composition analytics with predictive and operational applications creates a need for FAIR-compliant, interoperable, and representative data infrastructures that connect spectral, compositional, process, and supply-chain data across products and operational settings (Ahmad *et al.*, 2026; Shanmugapriya *et al.*, 2026; Zhang, 2026). Second, expanding predictive, operational, and personalized analytics requires explainable, robust, computationally efficient, and transferable models supported by common benchmarks and validation across products, populations, regions, and real-world contexts (Siripurapu *et al.*, 2026; Zhao *et al.*, 2026; Zhou *et al.*, 2026). Third, the emergence of semantic, generative, and knowledge-driven analytics creates opportunities for multimodal and hybrid architectures that combine data-driven learning with ontologies, knowledge graphs, and domain knowledge to connect compositional, operational, nutritional, and consumer evidence while improving semantic consistency and decision transparency (Gjorgjevikj *et al.*, 2026; Jasrotia and Singla, 2026; Sharma and Gupta, 2025). Finally, the increasing interaction among all four dimensions requires research on governance and implementation models that examine how data governance, algorithmic transparency, privacy, human oversight, and stakeholder participation can support trustworthy, equitable, and sustainable deployment across food-system settings (del Moral, 2026; Nguyen *et al.*, 2026; Singh *et al.*, 2026).

6. Limitations of the study

This study has several limitations. First, the corpus is limited by its reliance on Scopus and terminology-dependent retrieval. It may therefore exclude regional, non-English, practice-oriented, or differently labeled research relevant to Food Analytics. Iterative query refinement and eligibility screening reduced this risk but cannot guarantee complete coverage of the field. A single researcher validated eligibility decisions rather than multiple reviewers independently replicating them; despite computational support and predefined eligibility criteria, this may introduce selection bias through researcher judgment, particularly in borderline cases.

Second, emergence detection is sensitive to descriptor extraction, thesaurus construction, stopword selection, temporal windows, novelty and frequency thresholds, growth-ratio settings, and co-occurrence parameters. Alternative specifications could identify different emergent descriptors or alter cluster composition. Sensitivity analysis improved the stability

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and interpretability of the selected solution, but it does not eliminate methodological dependence on parameter choices.

Third, Louvain clustering produces overlapping thematic communities in an interdisciplinary field. Consolidating these clusters into broader research dimensions therefore required interpretive judgment based on relationships among descriptors and associated records. The proposed dimensions should be understood as an analytically grounded synthesis rather than as fixed or mutually exclusive domain boundaries.

Fourth, the 2026 publication window is incomplete; however, excluding these records would remove a substantial portion of the most recent evidence used to identify the research frontier, while annualization would require introducing synthetic descriptor counts. Accordingly, only observed 2026 records were retained, and the most recent emergence signals should be interpreted as provisional.

Finally, the study maps thematic emergence rather than evaluating the quality, effectiveness, maturity, or practical impact of individual technologies and applications. An emergent descriptor indicates recent novelty and growth, not proven superiority, sustained future importance, or successful implementation.

Future research could address these limitations through methodological extensions of the present design. Multi-database retrieval could assess the robustness of the identified frontier across bibliographic sources, provided that differences in indexing and metadata are appropriately harmonized. Expert-based validation could complement descriptor- and network-based interpretation of emerging themes, while longitudinal monitoring could determine whether the trajectories identified here persist, consolidate, transform, or decline over time. Future methodological development could also examine open and reproducible implementations of comparable bibliometric tech-mining workflows, facilitating independent replication and adaptation while preserving explicit documentation of descriptor normalization, emergence detection, and analytical decision rules.

7. Conclusions

This study addresses the gap between previous accounts of individual technologies and applications in food systems and the need to identify how recent developments combine to form the emerging research frontier of Food Analytics. Methodologically, it contributes an emergence-oriented analytical workflow that integrates descriptor enrichment and normalization with descriptor-level emergence detection before co-word network construction and community detection, thereby focusing the thematic analysis specifically on emergent rather than established research topics. Empirically, the analysis identified 111 emergent descriptors organized into five Louvain communities and subsequently interpreted as four higher-order dimensions: intelligent food quality and composition analytics; predictive and operational food-system analytics; personalized nutrition and consumer analytics; and semantic, generative, and knowledge-driven food analytics. Conceptually, these dimensions characterize Food Analytics not merely as an application domain of AI, but as an interdisciplinary socio-technical field in which food-science knowledge, information infrastructures, analytical intelligence, consumer contexts, and decision processes increasingly converge. Their interaction suggests a cumulative and integrative evolution from relatively specialized analytical applications toward a more interconnected research domain linking evidence generation, prediction, personalization, decision support, and knowledge integration. Translating this emerging structure into practice will depend on interoperable data, validation across real-world settings, and responsible governance. Finally, the detected emergence signals represent current novelty, persistence, and recent growth in the literature and should not be interpreted as evidence of demonstrated technological impact, effectiveness, or future importance.

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Acknowledgments
The authors acknowledges Universidad Nacional de Colombia for its institutional support and for providing an academic environment conductive to the development of this research.

Supplementary material
The supplementary material for this article can be found online.

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